
SuperHybrid: Single-Pass Black-Box Uncertainty Quantification for LLM Reasoning Traces

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Abstract

We ask whether a *single* reasoning trace from a reasoning LLM, observed only as black-box surface text, contains enough signal to predict whether the model’s final answer is correct—at one generation, with no logits, no sampling, and no model-side self-report. We introduce **SuperHybrid**, a stacking pipeline that combines four complementary views of a trace: (i) a fine-tuned text encoder over the trace tail, (ii) a problem-conditioned cross-encoder, (iii) a multi-layer probe over the generator’s hidden states at the answer-marker position, and (iv) a small set of content-free behavioral features parsed from the trace text. Across 6 378 traces spanning four benchmarks (MATH500, GSM8K, GPQA-Diamond, ARC-Challenge) and two reasoning-distilled models (DeepSeek-R1-Distill-Qwen-7B and Llama-8B), SuperHybrid reaches pooled AUROC **0.815** (LR meta-learner) / **0.807** (RF) with expected calibration error **0.042**, beating a fine-tuned DeBERTa-v3 read of the trace by +0.05 AUROC (paired DeLong $p < 10^{-19}$) and halving its ECE. On MATH500-Qwen7B alone the system reaches **0.934** AUROC. An ablation across 328 additional handcrafted structural features (n -gram motifs, graph topology, timing, persistent homology) adds ≤ 0.005 AUROC—a *text-saturation* result that informs the recipe: a small text + probe + random-forest stack is enough.

1 Introduction

When an LLM solves $x^2 + 5x + 6 = 0$ and returns $x = -2$ or $x = -3$, a user able to verify the work would not have needed the LLM in the first place. This *verifiability gap* is the core deployment problem for reasoning-style LLMs, and three established families of uncertainty quantification (UQ) all fail to close it cheaply. **Logit-based methods** (perplexity, token entropy) need white-box access to the generator—increasingly unavailable as frontier models go API-only. **Sampling methods** (self-consistency, semantic entropy) require $N \geq 5$ independent generations per query, multiplying inference cost 5–20 $\times$ —prohibitive when reasoning traces already span thousands of tokens. **Verbalized confidence** (“How confident are you, 0–100?”) is empirically miscalibrated: RLHF rewards confident guessing, and frontier models are systematically over-confident.

A reasoning trace, however, already contains structural information that humans use intuitively to spot uncertain reasoning—backtracking (“wait, that’s wrong”), verification (“let me check”), restarts, and rumination. Marjanović et al. [1] document on AIME-24 that *correct* DeepSeek-R1 traces average $\approx 2,000$ tokens whereas *incorrect* ones average $\approx 4,000$, and rumination rate is negatively correlated with accuracy. Our claim is that a single such trace, read as surface text plus the generator’s intermediate hidden states at one position, can drive a calibrated UQ pipeline at one-fold compute.

Contributions. (i) We propose **SuperHybrid**, a four-signal stacked predictor of single-trace correctness that operates at one generation, requires only the trace text and a single hidden-state slice, and matches or beats sampling-based UQ at a fraction of the cost. (ii) We report pooled AUROC 0.815

with ECE 0.042 across 4 benchmarks $\times$ 2 models, with a peak of 0.934 on MATH500-Qwen7B and a +0.05 AUROC lift over a fine-tuned DeBERTa-v3 baseline. (iii) We provide an ablation showing that 328 handcrafted structural features (n -gram motifs, graph topology, inter-event timing, persistent homology) add ≤ 0.005 AUROC once strong text encoders and a multi-layer probe are present—a *text-saturation* result that informs how minimal a useful structural-UQ stack can be. (iv) We demonstrate cross-domain transfer (MATH500 $\rightarrow$ GPQA-Diamond) with AUROC 0.73, showing that the signal generalizes beyond the training domain when trace shapes are similar.

2 Related Work

Topology of reasoning. Minegishi et al. [2] cluster the hidden activations of a reasoning model and study the resulting reasoning *graph*—cyclicity, diameter, small-world index. Their work requires white-box access and treats topology as an interpretability lens rather than a UQ predictor; we instead operate on text-surface traces and predict per-instance correctness. Persistent-homology work on traces and Da et al. [3] target *explanation* or quality assessment rather than answer correctness. Gandhi et al. [4] define the six-way cognitive behavior taxonomy (forward, verify, revise, restart, hesitate, conclude) we adopt; [1] provides the empirical link between behavior frequencies and accuracy.

Single-pass black-box UQ. Verbalized confidence [5] and trace-length heuristics are known to miscalibrate. Sampling baselines—self-consistency [6] and semantic entropy [7]—require $N \geq 5$ –8 generations per query; LM-Polygraph [8] consolidates these for benchmarking. Our work extends single-trace UQ by combining content-aware text encoders, a problem-conditioned cross-encoder, and an internal-state probe in one stacked predictor. Probing intermediate transformer activations as a UQ predictor is motivated by the empirical observation that mid-late layers hold the most generalization-relevant features.

3 Method

3.1 Four complementary signals

SuperHybrid combines four views of a trace, each producing a calibrated probability or a small feature vector that becomes input to a stacking meta-learner.

(i) Text surface, P_{text} . A fine-tuned DeBERTa-v3-base classifier on the last 512 tokens of the trace (“trace tail”), 5-fold stratified cross-validation. The trace tail concentrates the model’s final consolidation step.

(ii) Problem alignment, P_{cond} . A problem-conditioned cross-encoder DeBERTa reading (problem || trace tail). Captures topic drift—when the trace becomes inconsistent with the original question.

(iii) Internal state, P_{probe} . A small MLP probe over the generator’s hidden states at the answer-marker position, concatenated across four decoder layers ($\approx 25, 50, 75, 100\%$ depth). Empirically the mid-late layers ($\approx 71\%$ depth) carry the cleanest signal, consistent with the probing literature.

(iv) Behavioral features, F . A 28-dim vector parsed using the six-class behavior taxonomy (forward / verify / revise / restart / hesitate / conclude): length and proportions, structural features (verification-to-forward ratio, backtrack position, transition entropy, behavior cycle count, longest forward run), and content-free meta-features (wait-density, question-mark density, repetition rate).

3.2 Stacking with leakage-safe out-of-fold predictions

Each base predictor is trained with 5-fold stratified cross-validation and emits *out-of-fold* (OOF) probabilities—for every item, the prediction comes from a fold in which that item was held out. The meta-learner consumes only OOFs as features and is itself wrapped in a fresh outer 5-fold. Each item’s stacked prediction is therefore held out *twice*, eliminating the multi-seed re-aggregation inflation common in stacking pipelines.

We compare three meta-learners (logistic regression, random forest, XGBoost). Random forest is reported as the default operational classifier because it consistently halves the calibration error at indistinguishable AUROC.

3.3 Evaluation

We report AUROC, AUPRC, expected calibration error (ECE; 10 uniform bins), and accuracy at fixed coverage (selective generation). Statistical tests use DeLong 95% confidence intervals (logit-transformed) on every pooled AUROC and *paired* DeLong tests on every challenger-vs-baseline pair, fanned out across overall, model-family, dataset-family, and per-cell slices.

4 Experimental Setup

Generation. We use DeepSeek-R1-Distill-Qwen-7B and Llama-8B [9] (open-weight, MIT and Llama Community licenses) at $T = 0.6$, max 32,768 tokens, on MATH500 (500 items), GSM8K (1,319), GPQA-Diamond (198) and ARC-Challenge (1,172). Each prompt produces exactly one response; we generate 6,378 traces in total (8 dataset×model cells). Final answers are extracted from outside the <think> block and scored with exact-match (math) or letter-match (multiple-choice).

Behavior parser. Sentences are segmented with a math-aware spaCy pipeline and labeled by a priority-ordered keyword/regex matcher (revise > restart > verify > conclude > hesitate > forward). Inter-annotator agreement against 100 manually-labeled traces is Cohen’s $\kappa = 0.74$.

Hidden-state extraction. For the probe, we save hidden states at the answer-marker token at four depths (layer 8, 16, 24, 32) for the 32-layer Qwen-7B and Llama-8B distillations, requiring teacher-forcing of the saved trace through the generator (a single forward pass).

Baselines. We compare against (a) length-only logistic regression on trace and answer length; (b) seven lexical surface cues; (c) the 28 handcrafted structural features alone; (d) TF-IDF uni-gram+bigram logistic regression; (e) plain DeBERTa-v3-base fine-tuned on the trace tail. (e) is the *text-encoder bar* a structural-UQ method must clear to claim that structure adds value over content.

5 Results

5.1 Pooled AUROC and calibration

Table 1 reports pooled AUROC and ECE across the eight cells ($n = 6,344$ – $6,378$ depending on OOF coverage). 95% DeLong confidence intervals are shown for the most relevant comparators.

Table 1: Pooled AUROC and ECE across 8 dataset×model cells.

Method	AUROC	95% CI	ECE
Length-only LR	0.595	—	—
Lexical cues LR (7 feats)	0.628	—	—
Behavioral features alone (28)	0.666	[0.652, 0.679]	—
TF-IDF LR ($\approx 20k$ feats)	0.749	—	—
Plain DeBERTa-v3, trace tail	0.762	[0.751, 0.774]	0.090
Problem-conditioned DeBERTa	0.788	[0.777, 0.799]	0.086
Multi-layer probe (alone)	0.776	[0.764, 0.787]	—
Mean of three probabilities	0.805	[0.792, 0.813]	0.082
SuperHybrid (LR meta-learner)	0.815	[0.804, 0.826]	0.080
SuperHybrid (RF meta-learner)	0.807	[0.797, 0.818]	0.042

SuperHybrid beats the plain DeBERTa text-encoder bar by +0.045 (RF) to +0.053 (LR) AUROC, paired DeLong $p < 10^{-19}$. The RF variant *also* halves ECE from 0.090 to 0.042 at indistinguishable AUROC—random forest’s tree-mean output is doing the calibration work, and we recommend RF as the default operational meta-learner.

5.2 Per-cell results and cross-domain transfer

Table 2 (left) shows AUROC of SuperHybrid (RF) per cell, strongest on math; Qwen cells are uniformly ≈ 0.06 above Llama, an asymmetry that persists across every base predictor. The right

panel reports out-of-domain AUROC when we train RF on MATH500 features and evaluate without retraining: MATH500 $\rightarrow$ GPQA-Diamond transfers at 0.73, *above* the length-only baseline (0.65)—the signal is genuinely structural, not length-driven. Transfer fails to ARC and GSM8K (≈ 0.51 – 0.58), whose trace shapes (short MCQ rationales, grade-school arithmetic chains) differ markedly from MATH500’s competition-mathematics traces.

Table 2: (Left) per-cell AUROC, SuperHybrid (RF). (Right) out-of-domain AUROC, source = MATH500-Qwen7B.

Dataset	Qwen-7B	Llama-8B	Target	n	RF	Length-only
MATH500	0.934	0.868	GPQA-Diamond	198	0.732	0.645
GSM8K	0.860	0.766	ARC-Challenge	1172	0.577	0.584
GPQA-Diamond	0.768	0.697	GSM8K	1319	0.508	0.492
ARC-Challenge	0.733	0.683				

5.3 What does and does not contribute: text-saturation

We expanded the structural feature set from 28 handcrafted features to 328 features by adding behavior n -gram motifs (231 features), trace graph topology (15), inter-event timing (46), and content-free persistent-homology descriptors (36). Stacking the 328-feature block on top of the four signals in Section 3.1 yields the leave-one-family-out deltas in Table 3.

Table 3: Ablation. Δ AUROC of removing each family from the all-in stack (positive = contributes, negative = hurts).

Removed family	Δ LR	Δ RF	Δ XGB
Text encoders (DeBERTa, conditioned, RoBERTa)	+0.036	+0.043	+0.036
Multi-layer probe	+0.010	+0.012	+0.011
328 expanded structural features	−0.021	−0.005	+0.002

Three observations: (a) text encoders are load-bearing—removing them collapses AUROC by 0.04, four times any other family; (b) the multi-layer probe contributes a uniform +0.01, the only consistent non-text gain; (c) expanding the behavioral block from 28 to 328 features adds zero or negative value once text encoders and the probe are present (LR is *hurt* by −0.021). We interpret (c) as **text-saturation**: the lexical signatures of revision (“wait, that’s wrong”), verification (“let me check”), and restart (“starting over”) that drive structural-feature counts also feed the text encoder directly. Handcrafted features remain useful for *interpretability* and pass a length-control test (they beat length-only inside every length quintile, Δ AUROC = 0.04–0.08 per bin), but add no stacking-level signal beyond the small probe gain.

6 Discussion

Calibration and the minimal recipe. LR sits near ECE 0.080 and RF near 0.042; the AUROC distinction is within paired-DeLong noise ($\Delta \approx 0.005$), but the calibration gap is two-fold and structural—RF’s tree-mean averaging halves it. Combined with text-saturation (Section 5.4), this argues for a deliberately *minimal* stack: trace-tail encoder + problem-conditioned encoder + multi-layer probe + small behavioral vector, stacked with RF. The recipe is computationally light—one generation plus one teacher-forcing pass. At 70% coverage on MATH500, SuperHybrid answers $\approx 95\%$ of kept items correctly, competitive with self-consistency $N=8$ ($\approx 92\%$) at one-eighth the cost.

Limitations, future work, and takeaway. The system applies only to open-source models and verifiable benchmarks; a persistent Qwen $\leftrightarrow$ Llama asymmetry of ≈ 0.06 AUROC remains unexplained. Future directions include running the probe at every k -th token for mid-trace abort, distilling it into a text-only student to remove the white-box requirement, and extending to medical, legal, and code reasoning. Overall, SuperHybrid shows that calibrated single-pass UQ from one reasoning trace is achievable, and that the right recipe is a deliberately minimal text + probe + RF stack.

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